

Three homicides with darts tainted with succinylcholine: autopsy and toxicology

Jingjun Xing^{1,2} · Wenhe Li¹ · Fang Tong¹ · Yue Liang¹ · Guanglong He³ · Yiwu Zhou¹

Received: 21 December 2015 / Accepted: 29 April 2016 / Published online: 14 May 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract In emergency departments, intoxication with the muscle relaxant succinylcholine (SUX) often leads to a potentially lethal respiratory paralysis or other deleterious side effects. However, homicide cases with SUX poisoning are very rare because the toxic or lethal concentration ranges of SUX have not yet been determined. We described three uncommon homicide cases due to acute poisoning by darts contaminated with SUX. All the victims died quickly (less than 30 min) after being shot by an especially designed dart gun. Succinylmonocholine (SMC), a metabolite of SUX, was used as a marker to detect the latter. HPLC-MS/MS analysis demonstrated the presence of SUX in the droplet residues of the darts and SMC in the blood and urine in all cases. SMC concentrations of 0.45, 14.0, and 17.9 ng/ml were detected in the victims' blood and 259.0 ng/ml in the urine from the third case. The main pathological changes consisted of hemorrhage of the injured soft tissues, visceral congestion, severe pulmonary edema, and multifocal petechial hemorrhage of the heart and lungs. Taken together, the findings supported a diagnosis of fatal SUX poisoning. Furthermore, our study provided a reference for the lethal concentrations of SUX poisoning.

Keywords Succinylcholine · Muscle relaxant · Darts shooting · Fatal poisoning · Homicide

Introduction

Succinylcholine (SUX) is a nicotinic acetylcholine (ACh) receptor agonist, which structurally resembles the endogenous neurotransmitter ACh. It activates the postsynaptic ACh receptors in skeletal muscle to depolarize the endplate membrane resulting in paralysis. SUX is degraded by the specific acetylcholinesterase (EC 3.1.1.7) slowly, thus causing a prolonged depolarization [1, 2]. Based on its rapid onset, short duration of action, and rapid recovery, SUX is routinely used as a short-acting depolarizing muscle relaxant in anesthesia, during which the resulting respiratory paralysis is compensated by mechanical ventilation [1–4]. However, if the mechanical ventilation is not supported, a therapeutic dose of SUX can cause lethal apnea [1–7]. SUX is degraded within minutes by nonspecific cholinesterases (butyrylcholinesterase; EC 3.1.1.8), yielding succinylmonocholine (SMC), which is further metabolized into succinate and choline [1, 2, 4, 8]. SUX was detectable for up to 10 min and 2 h in blood and freshly secreted urine, respectively. However, detection of SMC in blood and freshly secreted urine appeared to be possible over a period of at least 8–24 h [2]. Therefore, SMC was considered to be the most promising target in SUX analysis in forensic science. In this study, SMC was thus used as the SUX marker.

In clinical and forensic practice, though fatal poisoning involving muscle relaxants [7, 9–11] have been reported, homicides using SUX are rare [12–14]. To our knowledge, there are no sufficient pathological data related to fatal SUX poisoning and very few data on the toxic and lethal SUX concentrations in humans have been published. Nowadays, despite being forbidden, the illegal possession of dart guns still exists

✉ Guanglong He
guanglong_he@163.com

✉ Yiwu Zhou
zhouyiwu@outlook.com

¹ Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, No. 13 Hangkong Road, Wuhan, Hubei 430030, China

² Medical School, Ningbo University, Ningbo, Zhejiang 315211, China

³ Institute of Forensic Science, Ministry of Public Security People's Republic of China, No. 17 Nanli Mulidi, Beijing 100038, China

in Mainland China. Some persons use dart guns contaminated with SUX to capture stray or domestic dogs to obtain dog meat and to sell for money. Because of this, forensic cases involving accidental or deliberate SUX application have occurred occasionally in China. Here, we report three uncommon homicide cases where the victims were shot with SUX-contaminated darts and died quickly. The SUX concentrations in the droplet residues of the darts and SMC concentrations in the blood and urine in these cases were also detected.

Case report

Case history

Case 1: A 46-year-old man was trying to stop a thief from abducting a dog when he was shot with a dart launched from a gun like emitter (Fig. 1a, b). He was hit on the inside of the right nipple and called for help while running but fell down suddenly about 10 m away. He was found dead when the emergency physicians arrived about 20 min later.

Case 2: A 63-year-old male was shot by a dart on the right elbow while quarreling with others. He ran away and collapsed; emergency physicians arrived 5 min later and declared him dead.

Case 3: A 28-year-old man found a masked peeper near his car while being intimate with his girlfriend at night. He complained of being shot by a dart like weapon in the back when he left the car to talk to the man. He went back to the car immediately and told his girlfriend that he had been shot and drove to the hospital. He stopped suddenly and collapsed on

the steering wheel after driving several hundreds meters. The victim died 15 min later when he was taken to the hospital.

Autopsy findings and histopathology examination

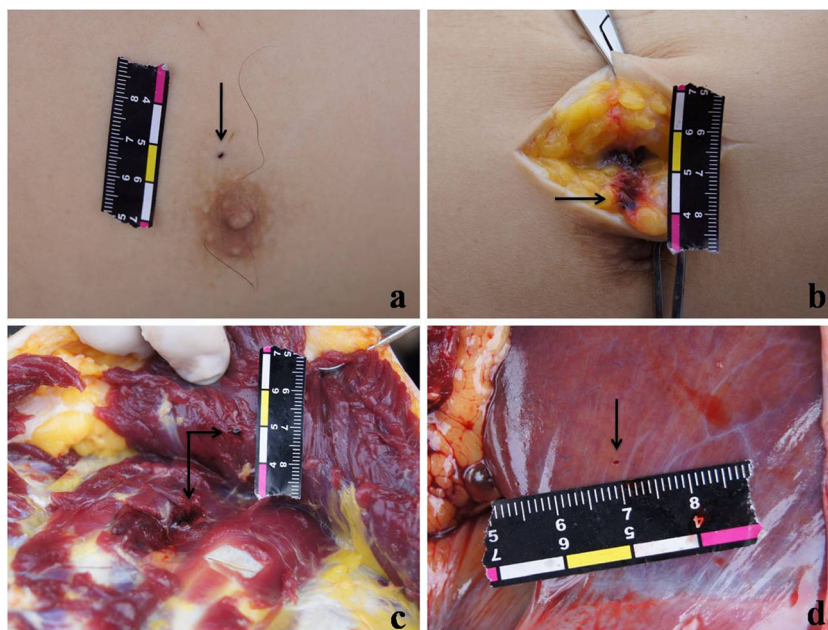
Forensic autopsies on these cases were performed 5, 2, and 12 h after death, respectively. External examinations of the bodies were unremarkable except for the needle holes with subcutaneous hemorrhage. In case 1, the stab extended obliquely through the chest wall into the abdomen and penetrated into the liver (Fig. 2). In case 2, the heart was unremarkable with minimal atherosclerosis and no evidence of myocardial necrosis. In case 3, though the injury history was clear; it was difficult to find the needle mark in the back near the spine. Focal hemorrhage in the anterior mediastinum and reddish foam in the lumen of trachea and primary bronchus were observed. The needle holes in all three cases were similar as shown in Fig. 2(a, b).

Macroscopical and histological examinations revealed multifocal petechial hemorrhage on the surface of the lungs and hearts in all cases (Fig. 3). Hyperaemia of the blood vessels and alveolar edema were noticed in all the lungs. The livers were congested in all the cases and with hemorrhage in the injured region in case 1. Hyaline thrombus was detected in some capillary vessels in the alveolar walls in case 2. Inflammatory cell infiltration in the subendocardial myocardium of the left ventricle and focal contraction band necrosis in the papillary muscles were found in case 3. Brain edema and some necrotic neurons (red neuron) in the brain were seen in all cases. In the other organs including the spleen, kidney, intestines, and lymph nodes, no lesions were observed except

Fig. 1 The gun-like emitter (a), (b) and the darts (c), (d). The volume of fluid tainted with SUX in the darts was approximately 0.2–0.3 ml (c), (d)



Fig. 2 Stab wound of the dart. **a** The needle hole located on the inside of the right nipple (*arrow*). **b** Section of the needle hole showing subcutaneous hemorrhage (*arrow*). **c** Hemorrhage in the corresponding intercostal muscles (*arrows*). **d** Needle hole in the liver (*arrow*)



for the varying degrees of congestion of the blood vessels. No other injuries or diseases were found in these cases.

Toxicology

Darts were found at the scene of the crime in all the three cases, and the volume of fluid in the darts was approximately 0.2–0.3 ml (Fig. 1c, d). Fresh autopsy specimens including blood, urine, and droplet residues in darts were sampled for toxicology analysis and were stored at -20° until analysis.

According to clinical histories and postmortem findings, a comprehensive drug and poison screenings including SUX analysis were performed. HPLC-MS/MS was used for SUX

analysis and performed as described previously [2]. Briefly, SUX was extracted from the droplet residues of the darts, and SMC was extracted from the plasma and urine using ion-pair extraction on polymeric reverse phase SPE (solid-phase extraction) cartridges. The extracts were analyzed by HPLC-MS/MS method in positive ion mode. Quantitation was obtained by an isotope dilution approach using a set of noninterconvertible internal standards (SUX- d_{18} and SMC- d_3). The results revealed the presence of SUX in the droplet residues of the darts and SMC in the blood and urine. SUX concentrations of 171.3, 197.8, and 256.8 mg/ml were detected in the droplet residues of the darts of these three cases, respectively. SMC concentrations of 0.45, 14.0, and 17.9 ng/ml were detected in the blood of these three cases, respectively. In the third case, SMC concentration of 259.0 ng/ml in the urine was also determined. Unfortunately, no urine specimens from case 1 and case 2 as well as other tissue specimens of all the cases were obtained. As an example, Fig. 4 shows the SMC detection in the blood from case 1. Other drugs and poisons, especially a cyanide, were all negative.

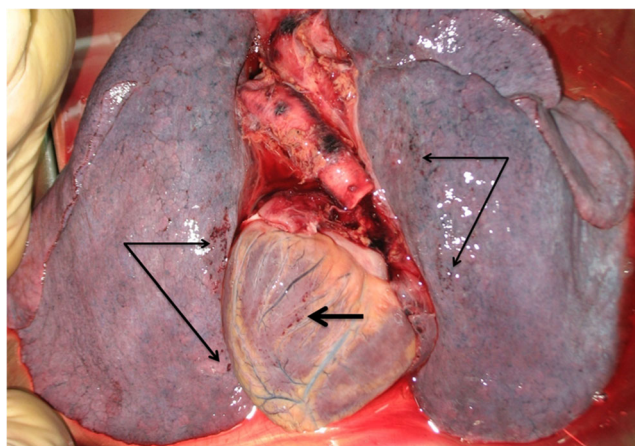
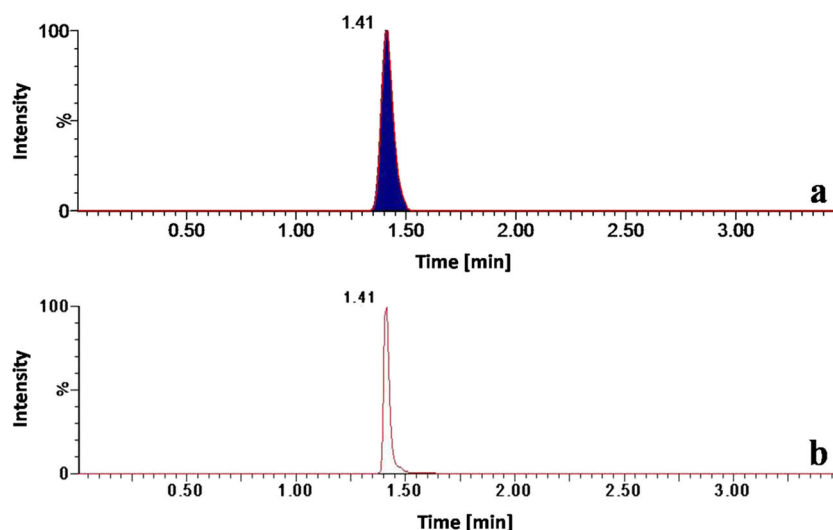


Fig. 3 Multifocal petechial hemorrhage on the surface of the lungs (*thin arrows*) and the heart (*thick arrow*) in case 3

Discussion

Criminal poisoning is uncommon, and the use of muscle relaxants for this purpose is even more rare. SUX can evoke respiratory paralysis and has many side effects, including hyperkalaemia leading to cardiovascular instability, bradycardia, neuroleptic malignant syndrome, and malignant hyperthermia [15]. SUX can also cause some side and toxic effects

Fig. 4 Detected succinylmonocholine (SMC) in the blood in case 1 (**a**) and in healthy blood with added standard SMC as a positive control (**b**)



in animals [16]. Intramuscular administration of a large dose of SUX in animals could cause prolonged respiratory paralysis, leading to lethal hypoxia [16]. In case 1, the victim was shot by a man trying to capture a domestic dog by shooting it with darts.

To our knowledge, there are very few pathological data related to fatal SUX poisoning in humans. In our study, no fatal injuries or diseases were found in any of the victims. The main pathological changes consisted of hemorrhage of the injured tissues, visceral congestion, severe pulmonary edema, and multifocal petechial hemorrhage of the heart and lungs, which indicated a symptom of apnea.

In addition, few data on the toxic and lethal SUX concentrations in humans have been published. It was reported that SMC concentrations of usually less than 50 ng/g in normal postmortem liver and kidney, while the brain (just like blood) can be considered SMC-free [17]. Concentrations of 3.7 µg/ml (SMC) and 8.1 µg/ml (SUX) in the urine have been reported in a fatal SUX intoxication case [4]. In our study, SUX was detected in the droplet residues of the darts, and SMC was detected in the blood and the urine. The SMC concentrations in the blood exhibited significant differences. Compared to the blood, the amount in the urine (259.0 ng/ml) was much higher. Even so, the urinary SMC concentration reported here was much lower than the previously reported (3.7 µg/ml) [4]. Studies have also revealed that the serum and urine are free of native SMC [17]. SUX and SMC are stable enough in the urine to allow detection during forensic analyses [2, 4]. These results demonstrated that urine was the proposed matrix of choice in forensic toxicological SUX analysis and consistent with previous studies [1, 2, 4, 17, 18]. The lowest SMC concentration in the blood (0.45 ng/ml) in case 1 and the urinary SMC concentration in case 3 indicated that the lethal SMC concentrations are even lower. Therapeutic dosages of SUX when applied intramuscular are about 3–4 mg/kg with the

maximum dose of 150 mg (1.5 mg/kg in intravenous application) [19]. The cases reported here confirmed that though being equivalent to or less than the therapeutic dosages, the dosages of SUX can cause acute fatality if the mechanical ventilation is not supported.

In summary, though neither the macroscopic nor microscopic findings could explain the cause of death in these three cases, appropriate clinical histories, autopsies and histopathological findings, demonstration of SUX in the droplet residues of the darts, and SMC in the autopsy specimens supported a diagnosis of acute SUX poisoning that lead to cardiopulmonary failure. Although homicide cases due to SUX poisoning are rather rare, the fatal cases reported here can help forensic pathologists and toxicologists confirm the cause of death in suspicious cases. The SMC concentrations reported here are also helpful to determine lethal SMC concentrations. In addition, SUX poisoning must be considered in any case that develops into respiratory failure or cardiac arrest quickly with a needle puncture. In these cases, it is very important to find out needle holes and prompt specimen sampling, and storage should be done.

Since SUX is not classified as a high toxic substance and poison in China, we suggest that a set of strict management guidelines of SUX as well as other muscle relaxants must be established to reduce such malignant events.

Acknowledgments This work was supported by the National Natural Science Foundation of China (Grant No. 81273336), Key Program of Natural Science Foundation of Hubei Province of China (No. 2013CFA073), and the Open Research Fund Program of the State Key Laboratory of Virology of China (No. 2013IOV006). We thank Dr. Basil Arif for editing the manuscript.

References

- Kuepper U, Musshoff F, Hilger RA, Herbstreit F, Madea B (2011) Pharmacokinetic properties of succinylmonocholine in surgical patients. *J Anal Toxicol* 35(5):302–311
- Kuepper U, Herbstreit F, Peters J, Madea B, Musshoff F (2012) Degradation and elimination of succinylcholine and succinylmonocholine and definition of their respective detection windows in blood and urine for forensic purposes. *Int J Legal Med* 126(2):259–269
- Combs JM, Combs GN (1994) A literature review of the newest muscle relaxant: ORG 9426. *CRNA* 5(3):104–112
- Kuepper U, Musshoff F, Madea B (2008) A fully validated isotope dilution HPLC-MS/MS method for the simultaneous determination of succinylcholine and succinylmonocholine in serum and urine samples. *J Mass Spectrom* 43(10):1344–1352
- Maltby JR (1975) Criminal poisoning with anaesthetic drugs: murder, manslaughter, or not guilty. *Forensic Sci* 6(1–2):91–108
- Strock D, Kuczkowski KM, Greenberg M (2004) Accidental administration of succinylcholine for the treatment of hypotension in a labouring parturient. *Can J Anaesth* 51(8):853–854
- Somogyi G, Varga M, Prokai L, Dinya Z, Buris L (1989) Drug identification problems in two suicides with neuromuscular blocking agents. *Forensic Sci Int* 43(3):257–266
- Lehmann H, Silk E (1953) Succinylmonocholine. *Br Med J* 1(4813):767–768
- Hendin H, Maltzberger JT, Haas AP (2003) A physician's suicide. *Am J Psychiatry* 160(12):2094–2097
- Martinez MA, Ballesteros S, Almarza E (2006) Anesthesiologist suicide with atracurium. *J Anal Toxicol* 30(2):120–124
- Sporkert F, Brunel C, Augsburger MP, Mangin P (2012) Fatal tolperisone poisoning: autopsy and toxicology findings in three suicide cases. *Forensic Sci Int* 215(1–3):101–104
- Maeda H, Fujita MQ, Zhu BL, Ishidam K, Oritani S, Tsuchihashi H, Nishikawa M, Izumi M, Matsumoto F (2000) A case of serial homicide by injection of succinylcholine. *Med Sci Law* 40(2):169–174
- Furbee RB (2006) Criminal poisoning: medical murderers. *Clin Lab Med* 26(1):255–273
- Johnstone RE, Katz RL, Stanley TH (2011) Homicides using muscle relaxants, opioids, and anesthetic drugs: anesthesiologist assistance in their investigation and prosecution. *Anesthesiology* 114(3):713–716
- Mirzakhani H, Welch CA, Eikermann M, Nozari A (2012) Neuromuscular blocking agents for electroconvulsive therapy: a systematic review. *Acta Anaesthesiol Scand* 56(1):3–16
- Oruç HH, Akkoç A, Alasonyalılar A, Aydın S (2009) Suspected succinylcholine poisoning in a German shepherd dog. *J Biol Environ Sci* 3(9):91–94
- Ballard KD, Vickery WE, Nguyen LT, Diamond FX, Rieders F (2006) An analytical strategy for quaternary ammonium neuromuscular blocking agents in a forensic setting using LC–MS/MS on a tandem quadrupole/time-of-flight instrument. *J Am Soc Mass Spectrom* 17(10):1456–1468
- Kuepper U, Musshoff F, Madea B (2011) Applicability of succinylmonocholine as a marker for succinylcholine administration-comparative analysis of samples from a fatal succinylcholine-intoxication versus postmortem control specimens. *Forensic Sci Int* 207(1–3):84–90
- Mason MA, Weant KA, Baker SN (2013) Rapid sequence intubation medication therapies: a review in light of recent drug shortages. *Adv Emerg Nurs J* 35(1):16–25